Scientific English

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M2 Marine Physics

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Lectures

Introduction:

- #1 A very Brief History of Scientific Writing
- #2 Scientific Writing today
- #3 What about AI?

Scientific writing:

- #4 Structure and content of a paper
- #5 Other formats (reports, proposals, etc.)

Effective scientific writing:

- #6–8 How to write more effectively (I, II, III)
- #9 Writing and revising a scientific paper

The peer-review process:

- #10 Reviewing a paper

Activities

- Read and discuss recent scientific articles
- Write a short article
- Review articles

Presentations and articles will be available at jgula.fr/SE

Part 3
What about AI?

***AI disclosure**: this presentation was prepared with the help of Claude Opus 5.5 (Anthropic), Oct. 2026, to search for sources, draft slides, and proofread. All content was checked and edited by J. Gula.*

~~Future~~ Present of scientific writing

Current revolution due to AI

- More and more authors use AI (e.g. Kobak et al, 2025)

- Is it OK to use AI to write science papers? Which use is allowed by journals, by UBO, by France?

AI in this course: not encouraged, but not ignored

- We will never encourage you to use AI to write your article: writing is thinking, and this course is about learning to write
- But AI cannot be ignored: more and more papers, and even peer reviews, are already written with it
- So let's look at it with open eyes: the numbers, the opinions, the rules, the risks and the costs
- If you use it anyway: follow the UBO rules, check everything, and declare it

AI and scientific writing: how widespread?

Kobak et al. (2025)?

AI and scientific writing: how widespread?

Kobak et al. (2025): 15.1 million PubMed abstracts (2010–2024)

- After ChatGPT (Nov. 2022), "excess words" appear abruptly: delves (×28 in 2024), underscores (×13.8), showcasing (×10.7), and more common ones: potential, findings, crucial…
- At least 13.5% of 2024 abstracts were processed with an LLM (≈200,000 papers per year): a lower bound
- Up to 40% in some subcorpora (e.g. 41% for computation papers from China); 7% for Nature, Science and Cell
- An effect on vocabulary larger than that of the Covid pandemic

How much of science is now written with AI? (2024–2026 studies)

- Liang et al. 2025 (Nature Human Behaviour): 1.1 M papers on arXiv, bioRxiv and in Nature journals (2020–Sept. 2024); LLM-modified content up to 22% in computer science, up to 9% in mathematics and Nature journals
- Kobak et al. 2025 (Science Advances): at least 13.5% of 2024 PubMed abstracts processed with LLMs
- Gray 2025 (arXiv:2512.01560): LLM tools likely contributed to more than 10% of all papers published in 2024 (full texts, Dimensions database)
- Siler 2026 (PNAS): 7.3 M articles (Elsevier, Frontiers, MDPI, PLoS), 2020–2025; 228 marker words flag ~12% of articles in 2023 and ~57% in 2025; higher for young for-profit publishers, lower-ranked institutions and regions far from English
- Caveat: word-based estimates measure LLM influence on the language, not authorship
- → Estimates differ by method and corpus, but all rise steeply after ChatGPT (Nov. 2022)

Excess vocabulary in your predecessors' articles

Analysis of LLM markers in articles written during previous years

Generate | + Code | + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline | ··· | cluster313 (Python 3.13.0)

LLM "excess vocabulary" in student articles (2015–2025)

Notebook version of `analyze.py`. Excess-vocabulary analysis of the Scientific English student articles, in the spirit of Kobak et al. (2025, *Science Advances*) and Q. Jamet's `excess_vocabulary.ipynb`. See `README.md` in this folder for the method, the results and the caveats.

Input: one plain-text file per article named `YYYY_NN.txt` (extracted from `all_articles/YYYY/*.pdf`). **Output:** only per-year aggregates (CSV files and figures). No individual article or student is identified, and these counts **must not be used to judge individual students**.

1. Settings

```
import re, glob, os, csv, math, random
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline

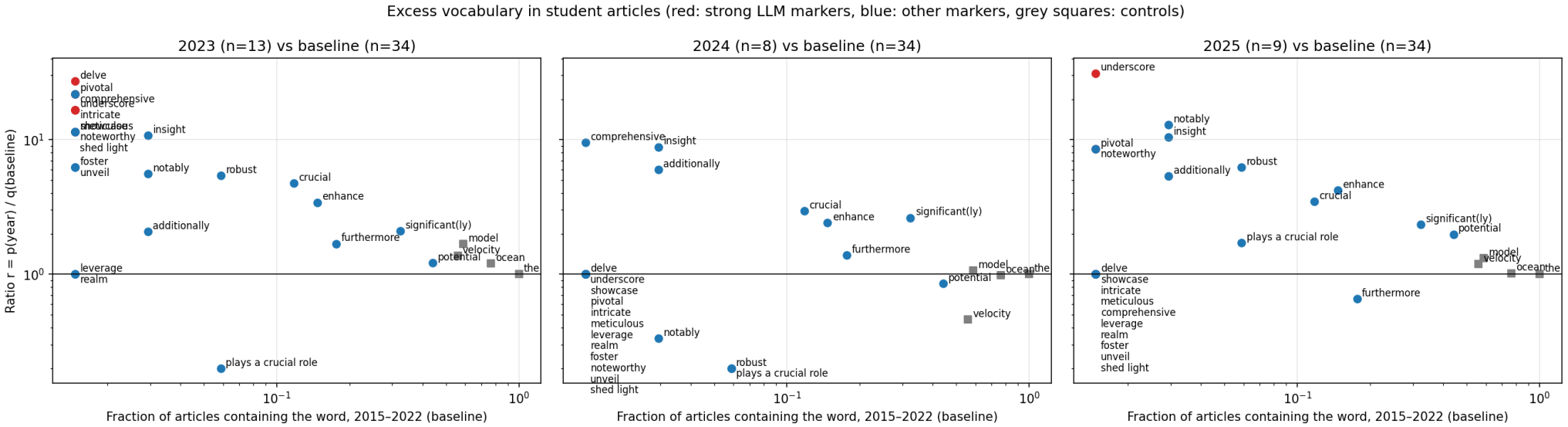
# Folder with the extracted texts (YYYY_NN.txt). The texts are not kept in the course folder (privacy).
TXT_DIR = '/Users/gula/Desktop/Work/Cours/M2/2627/SE/all_articles '
# Where CSVs and figures are written. Set to "." to overwrite the published results in this folder.
OUT = "notebook_output"
os.makedirs(OUT, exist_ok=True)
# Original PDFs, one sub-folder per year (read-only). Used for the overview table and the optional extraction.
ARTICLES_DIR = "../all_articles"
# Original PDFs, one sub-folder per year (read-only). Used for the overview table and the optional extraction.
ARTICLES_DIR = "../all_articles"
```

[4] ✓ 0.0s Python

Excess vocabulary in your predecessors' articles

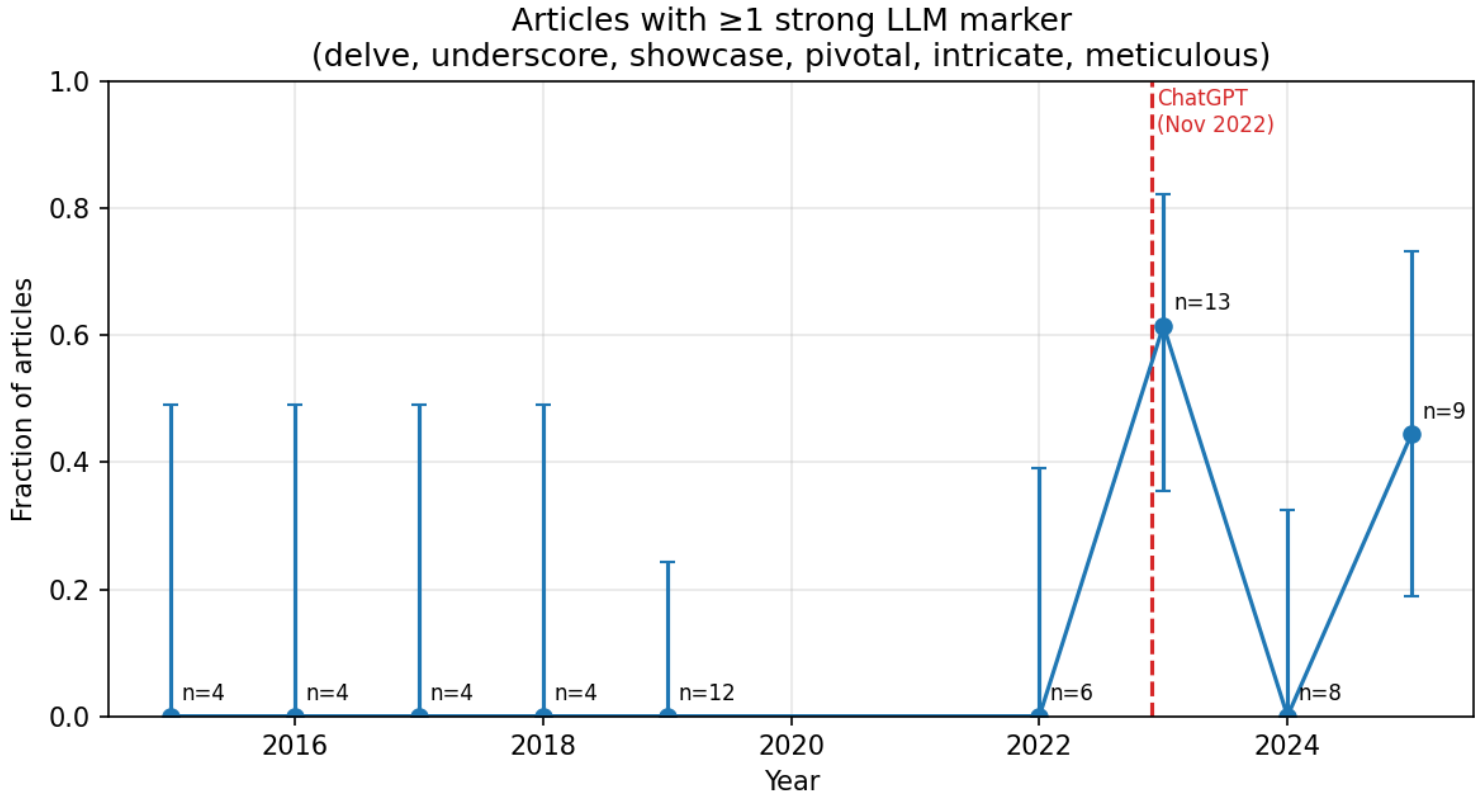
Analysis of LLM markers in articles written during previous years

Rare LLM markers (delve, notably, pivotal…) appeared in 2023



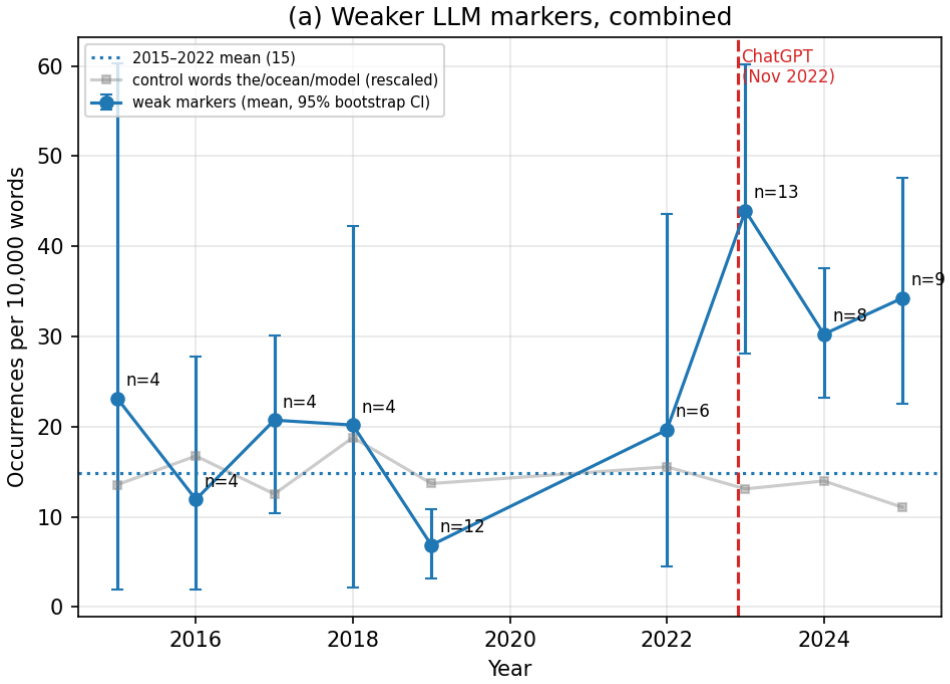
LLM vocabulary in M2 student articles, 2015–2025

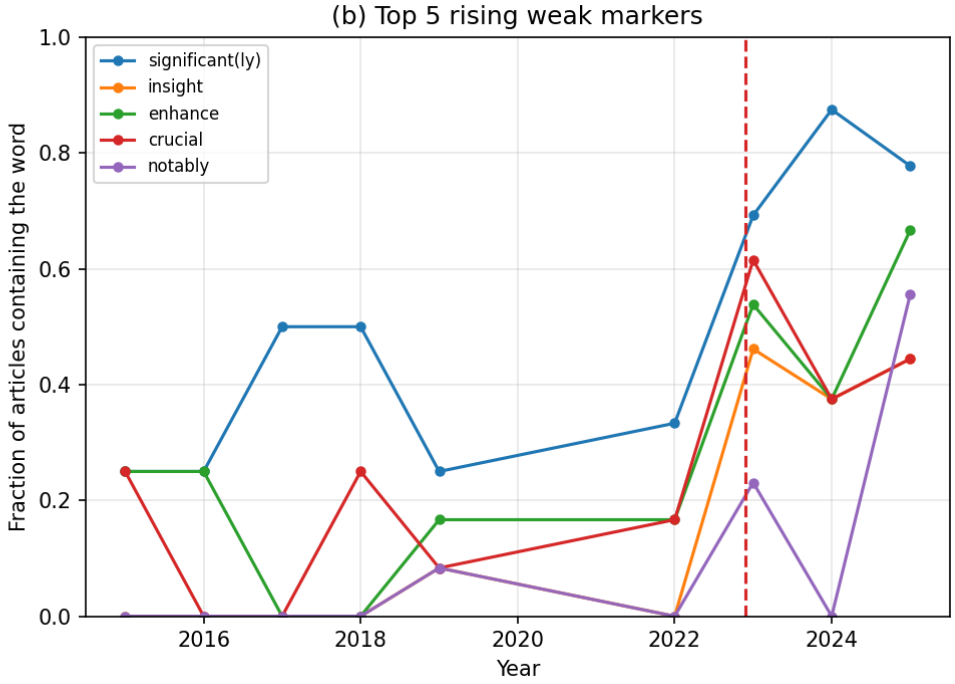
- Articles with ≥1 strong marker (delve, underscore, showcase, pivotal, intricate, meticulous): 0/34 in 2015–2022 → 8/13 in 2023 → 0/8 in 2024 → 4/9 in 2025
- 2024 dip, then back in 2025 (mainly "underscore"); common words stay high (e.g. "significant(ly)" 32% → 88% → 78%)
- Small samples; a style shift ≠ proof of AI authorship



Weaker markers: common words that also rose

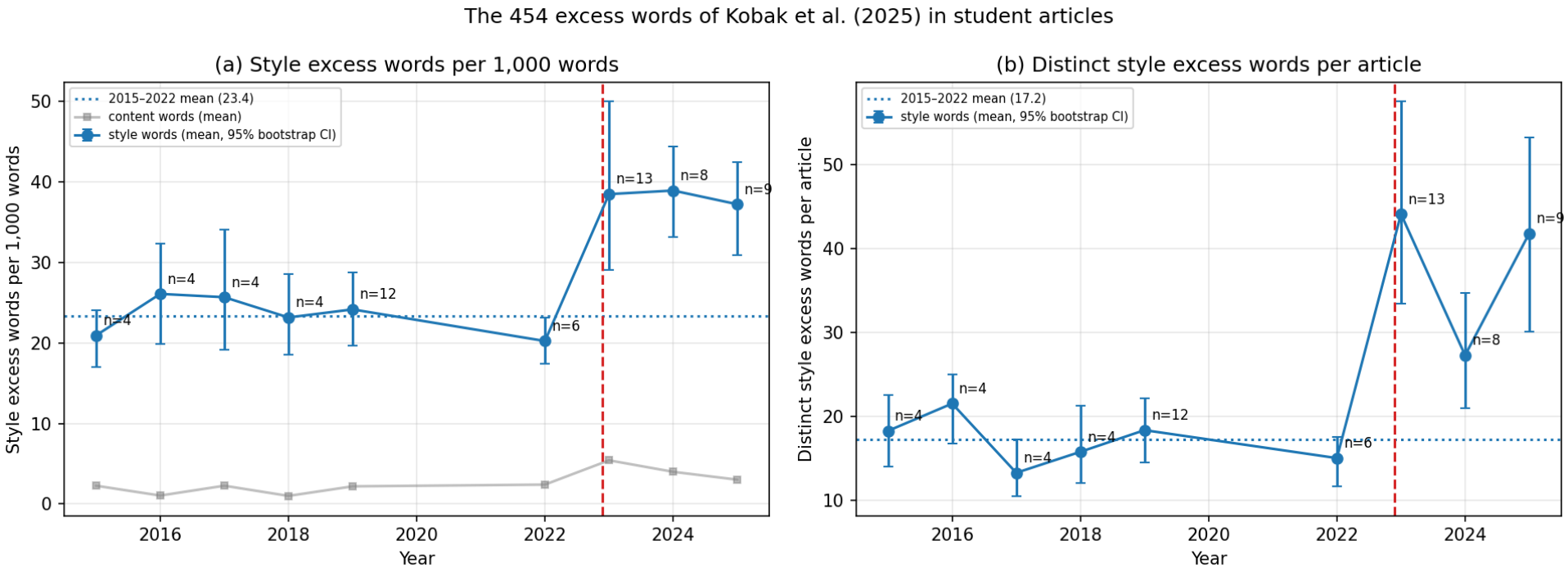
- Weak markers (crucial, significant(ly), insight, enhance, notably…): 15 per 10,000 words in 2015–2022 → 44 in 2023 → 30 in 2024 → 34 in 2025
- Largest rises in 2025: potential 44% → 89%, enhance 15% → 67%, notably 3% → 56% of articles; significant(ly) 32% → 78%
- These words are also common in human writing: the group trend matters, not single words





The 454 excess words of Kobak et al. (2025)

- Kobak et al.'s 454 excess words (379 style, 75 content): style words 23 per 1,000 words in 2015–2022 → 38–39 in 2023, 2024 and 2025
- The rise is significant every year (permutation test, $p \leq 0.0003$); content words change little
- Not our choice of words: a published list confirms the style shift, for the group, not for individuals



Poll 1: is it acceptable to use AI to…

- edit your paper?
- translate your paper?
- make summaries of other papers, to use in your own?
- write the first draft of your paper?
- do an initial peer review of a manuscript?
- assist your peer review by answering questions about a manuscript?

Answer for each: (a) appropriate (b) only if AI use is disclosed (c) only if the prompts are disclosed (d) never appropriate

Poll 2: which sections? And you?

A researcher writes one section of a paper with AI, then edits the result, without disclosing it. For each section: (a) appropriate (b) only with disclosure (c) never appropriate

- Abstract · Introduction · Methods · Results · Discussion / conclusion

Have you already used AI to… edit a text? translate it? summarise papers? write a first draft?

- (a) yes, without saying so (b) yes, and I said so (c) no, but I would (d) no, and I wouldn't

The Nature survey (2025): who was asked, and how

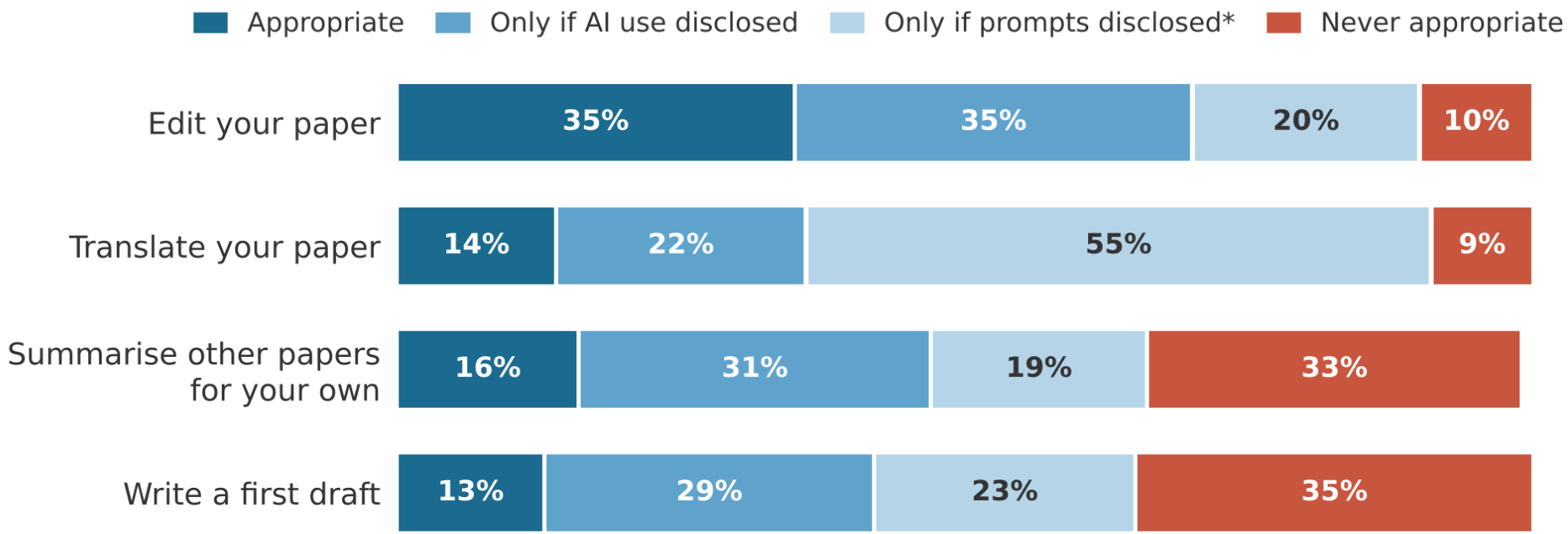
D. Kwon, "Scientists split on ethics of AI use", Nature 641, 574–578 (15 May 2025)

- 5,229 researchers answered in March 2025: authors of recent papers worldwide, a Springer Nature panel and Nature Briefing readers (21% USA, 10% India, 8% Germany; China under-represented)
- Scenarios about a fictional researcher, "Dr Bloggs", who uses AI without disclosing it: to edit, translate, summarise other papers, write a first draft or one section, or review a manuscript
- Two questions for each scenario: Is it acceptable? Have you done it yourself?
- Respondents were not told the journals' rules: the survey measures opinions, not policies
- Limits: anonymous volunteers, so response bias; not representative of all researchers

Article on the course web page: jgula.fr/SE/Nature_AI.pdf

Is it acceptable? Researchers' opinions

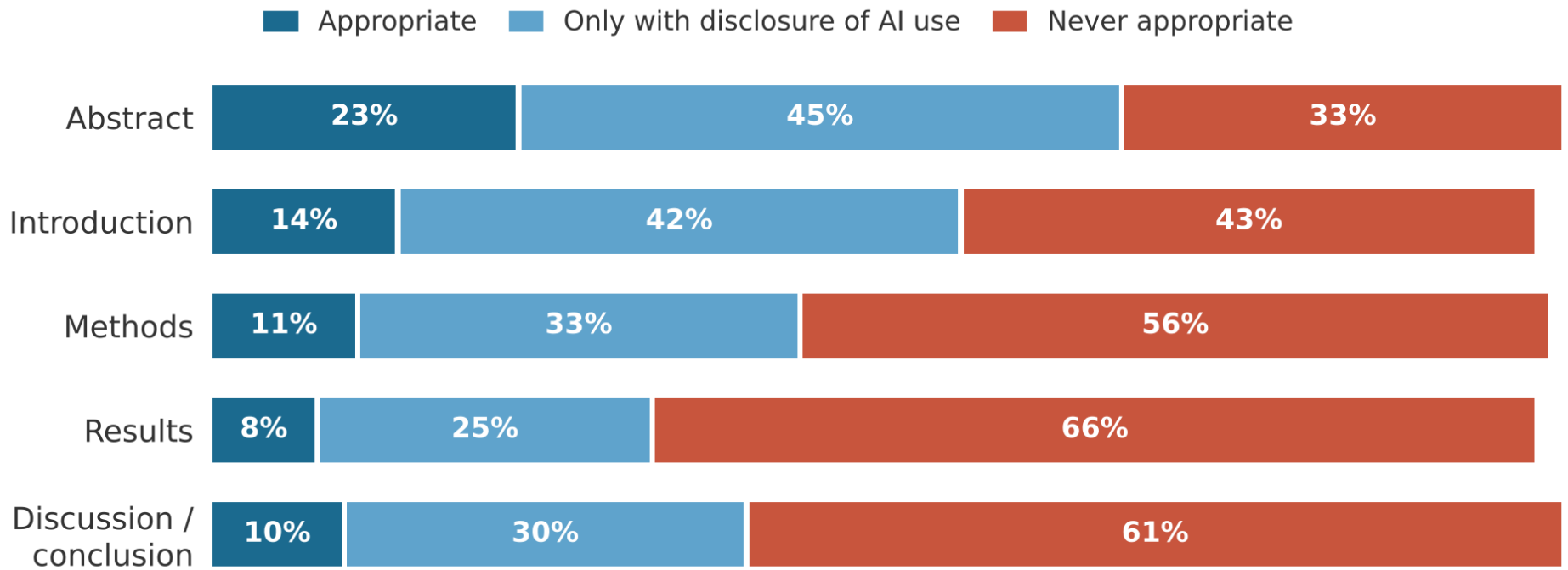
- Editing or translating your own paper: more than 90% find it acceptable, but many only with disclosure
- Writing a first draft: 65% accept it in some form; 35% say "never"
- Peer review: more than 60% reject an AI-written first review (a quarter of them for privacy reasons), but 57% accept AI assistance



Source: D. Kwon, Nature 641, 574–578 (2025); survey of 5,229 researchers. Percentages may not add to 100% (rounding).
*For translation: only if a fluent speaker checks it.

Which sections may AI write?

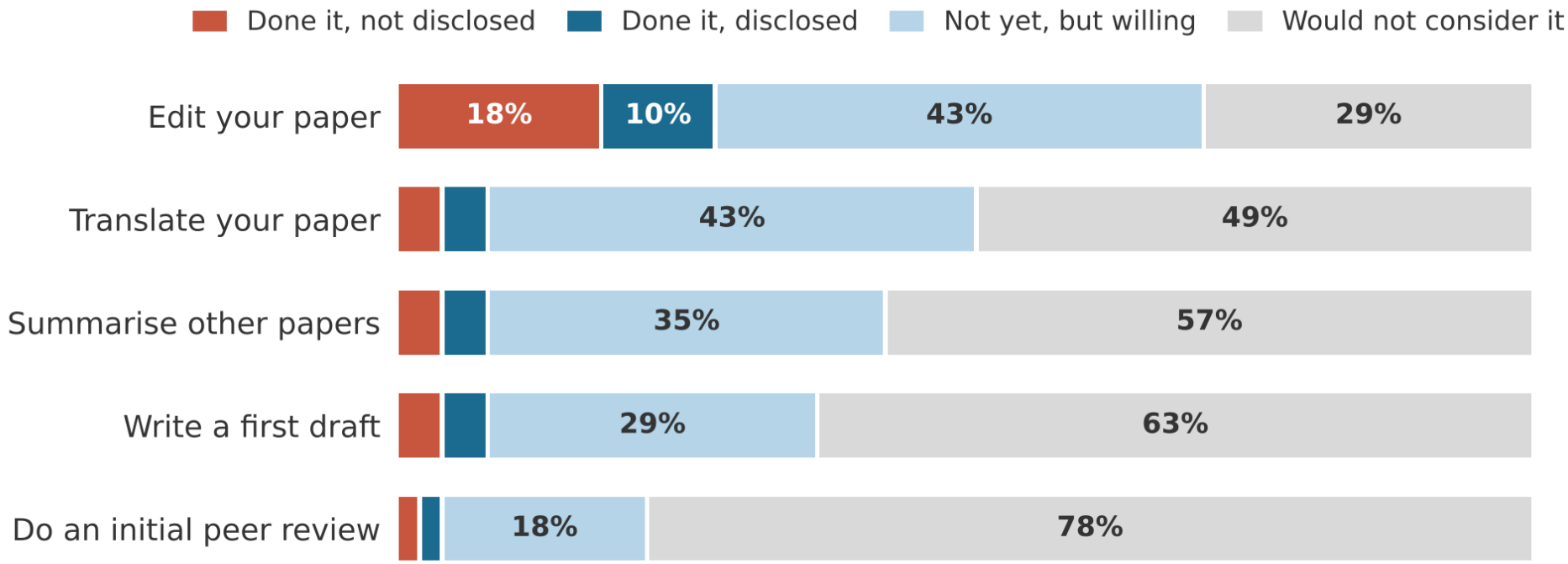
- The closer to the science, the stronger the rejection: “never appropriate” for 66% (Results), 61% (Discussion) and 56% (Methods)
- The abstract is the most accepted (33% “never”)… although it is the part that most people read



Scenario: "Dr Bloggs uses an AI tool to write a section of a paper, then edits the results, without disclosing it."
Source: D. Kwon, Nature 641, 574–578 (2025).

What do researchers actually do?

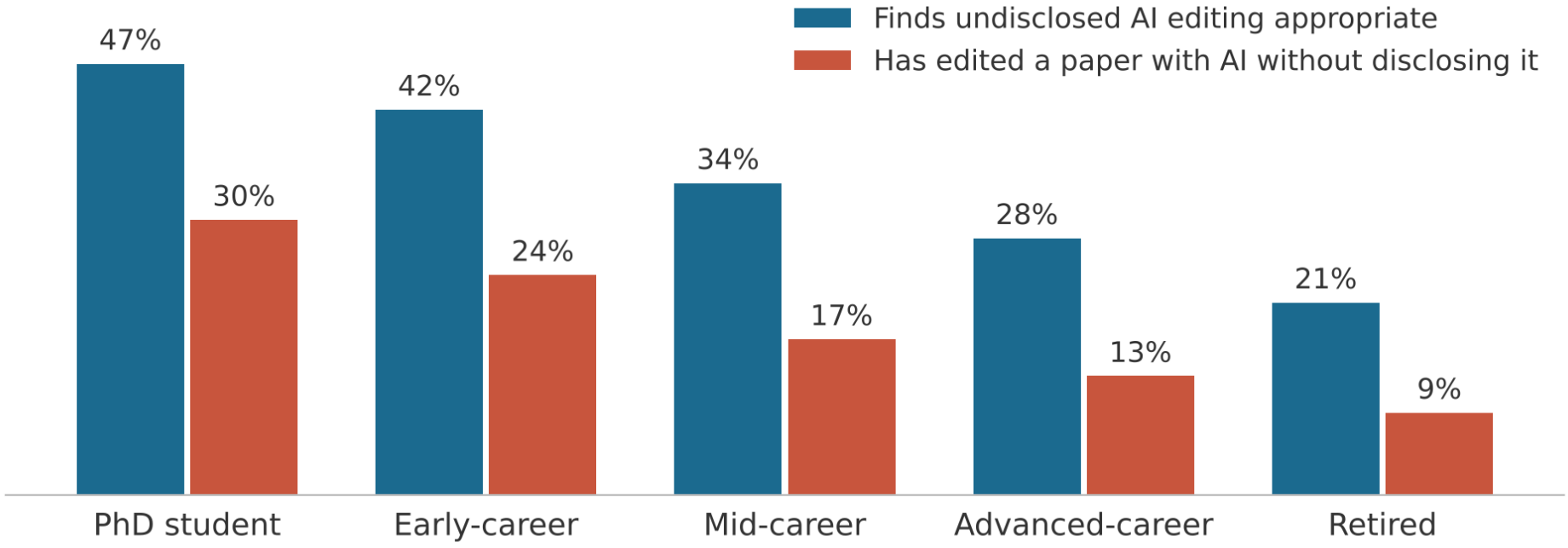
- “Great enthusiasm but low adoption”: 65% had used AI in none of these scenarios
- Editing is the most common use (28%), and most of these users did not disclose it (18% vs. 10%)
- About 8% used AI to translate, summarise or write a first draft; 4% for an initial peer review



Source: D. Kwon, Nature 641, 574–578 (2025); survey of 5,229 researchers. Percentages may not add to 100% (rounding).

Young researchers use AI more… and disclose it less

- PhD students are the most favourable to undisclosed AI editing (47%) and the most likely to have done it (30%)
- Opinions barely differ by country or field, but use is higher where English is not the first language
- Other surveys agree: 31% of ~2,000 medical researchers used AI to write or edit (Ng et al., Lancet Digit. Health 2025); Wiley (~5,000 researchers): 40% for translation, 38% for proofreading



Source: D. Kwon, Nature 641, 574–578 (2025) (scenario: Dr Bloggs edits his paper with AI and does not disclose it).

In their own words: a divided community

- “AI will be, if not already is, a norm just like using a calculator” (biomedical researcher, Bangkok)
- “It is pathetic cheating and fraud” (Earth-sciences researcher, Canada)
- “By using it, we rob ourselves of the opportunities to learn” (infectious-disease researcher, Cambridge)
- AI output can be “well-formulated crap”: false citations, inaccurate statements
- But AI can help level the playing field for researchers whose first language is not English
- “The norms of publication ethics have not changed: accurate reporting of what was done and why” (M. Wise, COPE)

Quotes from Kwon, Nature 641, 574–578 (2025)

AI and scientific writing: what journals require

- An AI tool cannot be an author: it cannot take responsibility for the work
- No AI-generated or AI-manipulated images or data
- Disclosure: improving your text (spelling, grammar, readability) usually needs none (STM 2023, Springer Nature); generating content does; Science asks for the prompts, JAMA for the tool and its maker
- Peer review: Elsevier and Science ban generative AI for reviewers; Wiley and Springer Nature allow limited use with disclosure, and never uploading the manuscript to a chatbot
- You remain fully accountable for every sentence, number and reference
- Policies differ and change (IOP Publishing dropped its disclosure mandate): read the author guidelines of your journal

Sources: Kwon, Nature 641, 574–578 (2025); publishers' author guidelines (e.g. AGU, Nature Portfolio, Science, Elsevier)

At UBO: the generative AI charter (2026)

Charte d'utilisation de l'IA générative, Université de Brest, approved by the Board of Directors on 5 Feb. 2026

- Applies to everyone: students, teachers, researchers, administrative and technical staff
- Legal frame: EU AI Act (Reg. 2024/1689), GDPR, intellectual-property code, UBO ethics charter and internal rules
- AI assists you; it must not replace the intellectual work expected from you
- Every AI output must be checked by a human: you are fully responsible for what you produce and share with it
- Use it sparingly: large energy footprint; a search engine or the library is often just as good
- Breaches can lead to disciplinary, administrative or even legal sanctions

Source: Charte IA UBO (2026), §1–5 and §8

UBO charter: what is forbidden, what is allowed

Forbidden

- Using AI instead of your own intellectual work, notably to produce academic content
- Using AI to answer an assessment: unless your teacher says otherwise, coursework produced by AI is cheating and can be sanctioned
- Putting personal, confidential or original research data into a tool not validated by UBO
- Getting round UBO's access restrictions; taking a decision based only on AI

Allowed, under human supervision

- Summarising documents you have the right to use; correcting spelling and syntax; rephrasing
- Organising your ideas; translating (with your teacher's agreement, unless translation is the exercise)

Source: Charte IA UBO (2026), §5 and §6

UBO charter: which tools, and how to declare their use

Tools

- “Sovereign” tools (data stay under French or EU public control): UBOtranscript for students and staff; the State’s Assistant IA and RAGaRenn for staff
- Not sovereign: ChatGPT, Claude, Copilot, Gemini, DeepSeek, Midjourney… and even Mistral (French, but its servers are run by US firms) → no personal or confidential data

Declaring: mandatory whenever AI helped to produce content

- Give the tool and its version, the date, what it was used for (ideas, plan, draft, proofreading, code…) and how you checked the result
- e.g. “We used ChatGPT (GPT-5 mini) on 12 Nov. 2026 to check the grammar of Sections 2–3; all changes were reviewed by the authors.”

Source: Charte IA UBO (2026), §7 and Annexes 1–2

The national framework: the State's AI guide (2026)

Guide d'usage de l'IA pour les agents publics de l'État (DINUM, DITP, DGAFP, V2, 2026):

a common reference for State employees; a ministry or institution charter, like UBO's, takes precedence

Five principles

- AI assists you, but you remain responsible for your decisions and productions
- Sensitive data only with the tools authorised by your administration
- Tell your readers when AI played a substantial role
- Use AI only when it is useful, watching for biases, copyright and environmental impact
- Train yourself to understand how AI works and what its limits are

Source: DINUM guide (2026), p. 5–6 and 14

State guide: which tool, when to declare, when to abstain

Three tiers of tools

- First, the tools of the administration (Assistant IA interministériel, Transcripts, Visio)
- Paid commercial licences provided by your employer: with care (EU hosting is not enough if the provider is subject to the US Cloud Act)
- Free commercial tools: only for information you could publish on a public website

Transparency and frugality

- Correcting, rephrasing or formatting a text you wrote → no mention needed; content produced by AI, even edited afterwards → “Partly generated by AI and checked by [name]”
- One text query ≈ 1 to 10 web searches; one image ≈ 60 times more: do not use AI to rewrite a three-line e-mail

Source: DINUM guide (2026), p. 10–12 and 18–20

Journals, UBO and the State: same core, different details

They all agree

- A human is responsible: AI is never an author, nor a decision-maker
- Check every output: facts, numbers, references
- No unpublished, personal or confidential data in commercial tools

They differ on disclosure

- Journals (STM, Springer Nature) and the State: simple editing of your own text needs no mention
- UBO: any AI help in producing content must be declared (tool, version, date, use)
- Science wants the prompts; for peer review, AI is often banned

When in doubt, “err on the side of disclosure” (A. Flanagin, JAMA, in Kwon 2025)

AI and scientific writing: risks

- Invented references and wrong facts ("hallucinations"): check every citation and every number
- Confidentiality: never upload unpublished data, or a manuscript you are reviewing, to a non-sovereign tool
- A homogenised, flowery style: "excess vocabulary" (delve, pivotal, showcase, underscore, intricate, notably, crucial…)
- e.g. a real 2023 abstract: "A comprehensive grasp of the intricate interplay between [...] and [...] is pivotal…"
- You learn less: writing is thinking

Sources: Kobak et al. (2025); Kwon (2025)

The environmental cost of AI

- Data centres: ~415 TWh in 2024 (1.5% of world electricity), ~945 TWh expected in 2030, a bit more than Japan uses today (IEA 2025)
- Training one large model (Mistral Large 2): 20,400 t CO_2e and 281,000 m³ of water
- Examples:
  - ~1 g CO_2e and 45 mL of water for 400 tokens (Mistral) with full life cycle (incl. server manufacturing), local electricity mix.
  - 0.03 g CO_2 and 0.26 mL for a median prompt (Google) data-centre use only, green-energy contracts (0.09 g without), cooling water only (~50 mL with power-plant water, S. Ren).
- One text query ≈ 1 to 10 web searches; one image ≈ 60 times more; reasoning models several tens of times more

- **Before using AI, ask: do I really need it? Could I do it another way?**

- Sources: IEA, Energy and AI (2025); Mistral AI with ADEME and Carbone 4 (2025); Google (Aug. 2025); DINUM guide (2026), p. 19

Detecting AI-written text

- Classifier detectors (Pangram, GPTZero, Turnitin) give each document a probability of being AI-written
- Pangram reports very low error rates (Emi & Spero, arXiv 2024) and estimated that ~21% of ICLR 2026 peer reviews were fully AI-generated (Nov. 2025)
- Limits: OpenAI withdrew its own classifier in July 2023 for low accuracy; detectors wrongly flag non-native English writing (61% of TOEFL essays on average, Liang et al. 2023); paraphrasing or light editing lowers scores
- A score is a probability, not proof: word counts (previous slides) work for groups, classifiers claim verdicts on single documents
-

Good use: AI as an editor, not a writer

- Write your own draft first: the ideas, the structure and the results must be yours
- Then ask for feedback rather than a rewrite: "point out unclear or ambiguous sentences", "check grammar"…
- Judge every suggestion with the principles of this course: *actions in verbs? characters as subjects? old → new information? needless words? no "excess vocabulary"? (to be seen later)*
- Reject changes that alter the meaning, add claims or add references
- Disclose your use as required by the journal and by the UBO charter

See also: Kwon (2025), Nature 641, 574–578

Finding the literature with AI: Undermind and Consensus

- **Search engines that read papers for you**
- Undermind (undermind.ai): describe your topic, answer its follow-up questions; it reads and ranks hundreds of papers, follows citations, and writes a report linked to each paper (a few minutes)
- Consensus (consensus.app): ask a question; it searches ~200 M papers (Semantic Scholar, OpenAlex…), ranks the top 20 and summarises what they say (Consensus Meter for yes/no questions)

- **Unlike ChatGPT, they search first, then summarise**
- The papers exist, but a summary can still misread them
- Gaps: paywalled full texts, the newest papers, books and reports, non-English work

- Sources: undermind.ai; consensus.app (Oct. 2026)

Nature Research Assistant (Springer Nature, beta)

- **What it does**
- Reading: summaries of a paper, a chat to ask questions about it, related papers
- Manuscript Adviser: upload your draft, get feedback on style, structure and substance, alternative titles and abstracts, places where a reference may be missing
- LLMs plus a knowledge graph of ~23 M papers (Springer Nature, Elsevier, open access; 2000–2025)

- **Comments**
- Run by a publisher: uploads are "not used to train" the model, but it is not a UBO-validated tool → no unpublished results or confidential data
- Using it does not change your chances of acceptance; check your journal's AI policy and disclose
- "AI can make mistakes": check every suggestion and every reference it proposes

- Source: natureresearchassistant.com, Help Centre (Oct. 2026)

Using AI search tools well

- Start from your own question and keywords; use the tool to find a few key ("seed") papers
- Open and read every paper you cite: never cite from an AI summary
- Cross-check with Google Scholar, Web of Science or Scopus, and follow the references of key papers
- Different tools find different papers: try more than one
- If a tool shaped your ideas or your plan, declare it (UBO charter, Annex 1)

- **Try it if you want: search your internship's topic in Google Scholar, Undermind and Consensus. What does each one find, or miss?**